

## List of standard symbols

|                                          |                                                                                         |
|------------------------------------------|-----------------------------------------------------------------------------------------|
| $\mathbb{R}$                             | Set of real numbers                                                                     |
| $\mathbb{N}$                             | Set of integers $\geq 0$                                                                |
| $\mathbb{Z}$                             | Set of all integers                                                                     |
| $\emptyset$                              | The empty set                                                                           |
| $ A $                                    | Cardinality of the set $A$                                                              |
| $\{x/x \text{ such that } \dots\}$       | Set of $x$ such that ...                                                                |
| $(\forall x)$                            | For every $x$                                                                           |
| $(\exists x)$                            | There is an $x$                                                                         |
| $a \in A$                                | $a$ is an element of the set $A$                                                        |
| $a \notin A$                             | $a$ is not an element of the set $A$                                                    |
| $A \cup B$                               | Union of $A$ and $B$                                                                    |
| $A \cap B$                               | Intersection of $A$ and $B$                                                             |
| $A - B$                                  | $A$ minus $B$ (elements of $A$ not belonging to $B$ )                                   |
| $A \subset B$                            | The set $A$ is a subset of set $B$                                                      |
| $A \not\subset B$                        | $A$ is not contained in $B$                                                             |
| $A \times B$                             | Cartesian product of $A$ by $B$<br>(set of pairs $(a,b)$ with $a \in A$ and $b \in B$ ) |
| $(1) \Rightarrow (2)$                    | Property (1) implies property (2)                                                       |
| $\binom{p}{q} = \frac{p!}{q!(p-q)!}$     | Binomial coefficient " $p$ choose $q$ "                                                 |
| $p \equiv q \pmod{k}$                    | The integer $p$ is congruent to $q$ modulo $k$                                          |
| $\left\lfloor \frac{p}{q} \right\rfloor$ | Integral part of $\frac{p}{q}$ (largest integer $\leq \frac{p}{q}$ )                    |
| $\left\lceil \frac{p}{q} \right\rceil^*$ | Smallest integer $\geq \frac{p}{q}$                                                     |
| $((a_j^i))$                              | Matrix in which the element in the $i$ th row and $j$ th column is $a_j^i$              |
| $\det((a_j^i))$                          | Determinant                                                                             |
| $\log p$                                 | Neperian (natural) logarithm                                                            |

For the notations specific to graphs, see the reference: *Graphs* (C. Berge, *Graphs*, North Holland, 1985).